Cover Letter

To: Editorial Office, *npj Quantum Information* (Nature Portfolio)
From: Aleaddin Özer (Corresponding Author) · ORCID 0000-0001-9389-5357
E-mail: ozer@e2esolutions.tech
Date: 2 June 2026

Dear Editors,

Please consider our manuscript, *"Three Axes of Quantum Risk: A Unified Observability Model for PQC, QKD, and Crypto-Agility,"* as an Original Research Article for *npj Quantum Information*.

The work started from a practical frustration. NIST has finalised its post-quantum standards and the NSA has put calendar deadlines on the migration, so operators are now being asked to inventory their cryptography and act on it. The tools they have for that job report a single bit per asset: PQ-resistant, or not. In practice that bit hides most of what an operator needs to decide where to spend a migration budget. Two TLS terminators can both read "classical," yet one is a configuration change away from a hybrid key exchange while the other is locked to a FIPS firmware build that will not move for eighteen months. And the single-bit view has nothing at all to say about quantum key distribution, which protects the channel rather than the primitive and is being deployed on exactly the high-assurance links where the migration matters most.

We make the case that quantum-risk posture is a three-axis measurement, not a one-axis label. The axes are algorithmic resistance, the protection on the channel that carries the key material, and how quickly the primitive can actually be replaced. We combine them with the operator's own deadline into a single score and — this is the part we care about — we build the instrumentation that observes all three on real systems, and we run it.

We think the channel axis is where the manuscript speaks most directly to this journal. QKD is usually studied as a physical-layer or protocol question; here it appears as a measured property of a deployed link, attributed per session, and validated on an open ETSI GS QKD 014 emulator that reproduces the link and key-management failure modes a real deployment would see. We drove thirteen such failures and the channel-state classification was correct on every one, including the case where one key-management endpoint fails while its healthy pair keeps delivering keys — a situation link-level telemetry alone cannot resolve. For a readership that has built the QKD systems being deployed, we hope the question "how would an operator actually observe this in production?" is a useful one to put on the table.

The empirical results are deliberately modest and fully reproducible. On the Tranco top-1000 we measured 43.8% of responsive hosts negotiating a hybrid PQ key exchange when offered one. On an eleven-project crypto-agility pilot we reached 91% agreement with hand-graded ground truth. None of the studies needs QKD hardware or a commercial scanner; they run on one Linux host. Everything — the schema, the implementation, the emulator and its replay corpus, the agility rule pack, the hand-graded data, and the Tranco snapshot identifier — is released under the MIT License at https://github.com/e2esolutions-tech/sezar.

On open-access fees: both authors are self-funding this work, with no grant or industrial sponsor behind it. We would be grateful for any waiver or discount the journal can extend, and we are submitting in the hope of publishing with *npj Quantum Information* either way.

The usual declarations: the manuscript is original and not under consideration elsewhere; both authors approved it and their contributions are stated in the back matter; we disclose one competing interest — the corresponding author is Chief System Engineer of E2E Solutions, which develops the open-source Sezar implementation described here, though the company earns no direct revenue from publication; the work was self-funded; and the data and code availability statement points to the open repository above. We suggest five reviewers in the accompanying file, three of them first authors on work we cite, none with any co-author or advisory tie to us. On acceptance we would publish under CC-BY 4.0.

Thank you for your time and for considering the manuscript.

Yours sincerely,

Aleaddin Özer
E2E Solutions · Hacettepe University
ORCID 0000-0001-9389-5357

Murat Aydos
Associate Professor, Hacettepe University
ORCID 0000-0002-7570-9204